

CSE 4412: Introduction to Quantum Computing - Week 2 Notes

D-separation and Conditional Independence

D-separation is a graphical criterion used to determine conditional independence in Bayesian networks. It allows us to identify whether two sets of variables are independent given a third set of variables.

Bayes ball algorithm

We can define a set of balls, and then we try to move them without them colliding or each other if that move is permitted.

WE cannot write an undirected graphical model such that this is the only conditional independence statement, the only conditional independencies implied here are below. We cannot write a directed graphical model such that these are the only

definition, a clique is a completely connected subgraph. A maximal clique is a clique that cannot be extended by including an adjacent vertex. A maximum clique is a clique of the largest possible size in a given graph.